



Flapping of Multifold Heliospheric Current Sheet Observed by Parker Solar Probe

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Abstract

The heliospheric current sheet (HCS) represents the most expansive current sheet within the heliosphere. However, there has been a lack of specific analysis regarding the potential folding and flapping of the HCS. The Parker Solar Probe (PSP) offers a unique opportunity to investigate this phenomenon because it can traverse the HCS multiple times during its perihelion encounters with the Sun. During the perihelion of Encounter 7, PSP successively crossed the HCS five times. Utilizing the minimum variance analysis method and discontinuity type analysis on the in situ measurements, we observed that the HCS exhibits a multifolded structure at small scales, coupled with an upward and downward flapping motion. The flapping velocities are estimated to range from several to hundreds of kilometers per second. By employing a two-step ballistic backmapping of the magnetic field in conjunction with observations from the Atmospheric Imaging Assembly on the Solar Dynamics Observatory, we deduced that the flapping motion may be instigated by a solar flare eruption at the source region. Furthermore, we confirmed that the multiple folds are not merely localized manifestations of the Kelvin–Helmholtz instability. Our research indicates that the HCS is, in reality, a dynamic 3D structure characterized by a complex small-scale morphology and its movements are intricately linked to the activities in the source region, revealing a level of complexity that surpasses previous understandings.

Unified Astronomy Thesaurus concepts: [Slow solar wind \(1873\)](#)

1. Introduction

In recent years, more and more studies have shown interest in the flapping motions of the magnetotail current sheet (CS) of planets, including Earth (V. Sergeev et al. 1998, 2003; C. Shen et al. 2008; A. Runov et al. 2009; Z. Rong et al. 2010), Venus (Z. Rong et al. 2015b), Mars (G. A. DiBraccio et al. 2017; C. Zhang et al. 2023), Mercury (C. Zhang et al. 2020), and Saturn and Jupiter (M. Volwerk et al. 2013). For the flapping of Earth's magnetotail CS, it is estimated that the velocity amplitude ranges from tens to hundreds of kilometers per second (V. Sergeev et al. 1998), and the displacement amplitude ranges from 1 to 3 R_E (A. Runov et al. 2009). Z. Rong et al. (2015a) suggested that flapping motions can generally be divided into two types, that is, the propagating kink-like type and the nonpropagating steady type. The kink-like type is characterized by flapping waves propagating along the CS, whereas the steady flapping corresponds to up-and-down flapping without propagation. Based on multipoint observations, the flapping type and flapping velocity can be easily determined.

Although the mechanism underlying magnetotail CS flapping remains unclear, numerous models and theories have been proposed and debated. Studies suggest that flapping may result from Alfvénic fluctuations of the interplanetary magnetic field (T. Toichi & T. Miyazaki 1976), the drift-kink mode of CS instability (H. Karimabadi et al. 2003; M. Sitnov et al. 2006), magnetohydrodynamic ballooning-type waves (I. Golovchanskaya & Y. P. Maltsev 2005), or hemispheric

force imbalance across the CS (H. Malova et al. 2007). N. Erkaev et al. (2007, 2009) proposed a prominent idea that the existence of two gradients of magnetic field, $\nabla_n B_\tau$ and $\nabla_\tau B_n$ relative to the CS, could give rise to the magnetic double-gradient instability, which is responsible for the flapping waves. This theory has been shown to be more consistent with observations in Earth's magnetotail, though there remains a slight discrepancy in the order of magnitude of the frequency (C. Forsyth et al. 2009).

As for the morphology, the foldings of the CS have been predicted in theory (S. Suess & E. Hildner 1985), reproduced in simulation (V. Pizzo 1994; P. Riley et al. 2002), and detected in observation (N. Crooker et al. 1993; M. Neugebauer 2008; C. Foullon et al. 2009; J. Peng et al. 2019). It is thought that the distortion of the CS is formed due to velocity inhomogeneities on both sides (S. Suess & E. Hildner 1985), especially the Kelvin–Helmholtz instability (KHI; M. Hesse et al. 1998; M. Hesse & J. Birn 2000). Furthermore, G. Lapenta & J. Brackbill (2002) noted that the required velocity shear for KHI could be one of the results of the nonlinear evolution of lower hybrid drift instability in a Harris CS.

The flapping motion, including steady and kink-like types, manifests as substantial velocity and spatial displacement of the CS in the inertial frame. In contrast, folds describe the intrinsic morphological features of the CS, which persist even in the absence of any dynamic motion. We emphasize here that these two concepts address two independent aspects, dynamics and morphology, of the CS. They are distinct phenomena but can coexist simultaneously.

The heliospheric current sheet (HCS) is the boundary between open magnetic field lines of opposite polarities, marking the magnetic equator of the global heliosphere. The thickness of the HCS is typically 10,000 km and increases



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approximately proportional to the heliocentric distance (E. J. Smith 2001). On a large scale, the HCS appears wavy and resembles a “ballerina skirt” in three dimensions, dividing the heliospheric magnetic field (HMF) into several sectors (H. Alfvén 1977; Y.-M. Liu et al. 2014; J. Huang et al. 2016). J. Gosling et al. (2006) suggested that crossings of the HCS are characterized by simultaneous reversals in the polarity of the HMF and the polarity of the strahl flow, beams of suprathermal electrons (H. Rosenbauer et al. 1977).

HCS is an important physical structure whose location and shape can significantly affect the propagation and geoeffectiveness of interplanetary shocks upstream of the Earth (Y. Xie et al. 2006; X. Zhao et al. 2007). In addition, the fine-scale structures within the HCS might affect the turbulent energy dissipation and thereby contribute to the heating process of the solar wind. However, as the most extensive CS in the heliosphere, there is no detailed conclusion about whether and how the HCS flaps. Meanwhile, the investigations about the folds of HCS mentioned in the previous paragraph are mostly limited to spatial scale over tens of Carrington longitudes and regions beyond 1 au.

Since the launch of the Parker Solar Probe (PSP) in 2018, an increasing number of mysteries regarding the inner HCS have been revealed (B. Lavraud et al. 2020; A. Szabo et al. 2020). Taking this opportunity, we aim to address the questions concerning the folding structure and flapping motion of near-Sun HCS. Through analyses of in situ measurements of PSP, we confirm that the HCS exhibits multiple small-scale folds and flapping motion with the CS normal velocity on the order of several to hundreds of kilometers per second. By back-mapping the magnetic field to its source region, we find, based on observations from the Atmospheric Imaging Assembly (AIA; J. R. Lemen et al. 2011), that the flapping of the HCS results from a solar flare that erupted several hours earlier. Furthermore, the observed velocity shear is insufficient to trigger the KHI and form the multifold morphology further. Therefore, we point out that HCS is neither smooth nor static, but rather a complex structure profoundly influenced by solar activities.

This work is structured as follows. First, Section 2 presents an overview of the in situ observation event by PSP. Next, Section 3 sequentially demonstrates that the HCS exhibits small-scale multifold structures and concurrent flapping motion, with the flapping velocity in the CS normal direction also provided herein. Subsequently, we infer in Section 4.1 that the multiple folds are not locally generated by the KHI, whereas Section 4.2 shows that the flapping motion originates from a solar flare in the source region. Finally, Section 5 summarizes our work and discusses its contributions to understanding the HCS and its relationship with the dynamics of solar sources.

2. Event Overview

Near its perihelion of Encounter 7, PSP traversed the HCS five times consecutively between 13:00:00 and 15:00:00 on 2021 January 17 (Figure 1). According to the specific variations of the magnetic field (mainly B_r) and the pitch angle distribution of strahl electrons of 314 eV, we identify the series of these five crossings as: complete (traversing from one side to the other), partial in (traveling into the HCS), partial out (returning to the same side), complete and another complete crossing.

Consecutive CS crossings may occur in four possible scenarios: (a) a static folded CS, (b) steady flapping of an unfolded CS, (c) kink-like flapping of a folded CS, and (d) steady flapping of folded CS. These scenarios are shown in the inertial reference frame in Figure 2. Considering both flapping motion and multifold structures, we find that the wavy CS morphology observed by the spacecraft can originate from three scenarios: (1) intrinsic folds without flapping (Figure 2(a)), (2) intrinsic folds coupled with steady flapping (Figure 2(d)), and (3) kink-like flapping motion (Figure 2(c)). To determine which scenario applies to our event, we need the normal direction and magnitude of the flapping velocity for each crossing—analysis of which will be presented in the next section.

3. Observations: Morphology and Dynamics

For each of these five crossings, we apply the minimum variation analysis (MVA) method to obtain the local CS (LMN) coordinates, where e_L represents the direction of the shearing component, e_N represents the normal direction of the CS, and e_M completes the right-handed system with $e_L \times e_M = e_N$. It is suggested that the quality of the coordinates resulting from MVA can be measured by the ratios of eigenvalues, λ_1/λ_2 and λ_2/λ_3 . However, no standard has been acknowledged so far (Z. Vörös et al. 2017). Taking the frequently used threshold of $\lambda_2/\lambda_3 > 3$ (A. Runov et al. 2009; C. Zhang et al. 2020), we select the suitable time interval to employ MVA, and determine all LMN coordinates well enough. Since the normal direction of the HCS derived from the MVA method has a 180° ambiguity, corresponding to two possible solutions. To resolve this ambiguity, we define the HCS normal as the direction pointing from negative B_r to positive B_r , as listed in Table 1.

The normal directions of these crossings, and angles between their neighbors, $\langle e_{N,i}, e_{N,i+1} \rangle$, are also listed in Table 1. We find the normal directions of HCS are sloping in the T or N direction of the RTN coordinate, in line with the fact that HCS is roughly extending outward from its root at the streamer cusp. Notice that the angles between adjacent normals are all greater than 45° , and $\langle e_{N,1}, e_{N,2} \rangle$ and $\langle e_{N,4}, e_{N,5} \rangle$ even reaches over 90° . This indicates successive sharp changes in the normal direction of HCS within less than 2 hr; therefore, we point out that HCS is a highly bent structure with multiple folds at small scales.

Using multipoint analyses of the Cluster or THEMIS data, the velocity vector of magnetotail CS flapping has been determined (V. Sergeev et al. 2003, 2004; A. Petrukovich et al. 2006; A. Runov et al. 2009). It is also worth noting that for planetary space exploration, when only a single spacecraft is available at local space and time, it is difficult to directly calculate the propagation velocity of the flapping CS. To address this challenge, Z. Rong et al. (2015a) and C. Zhang et al. (2020) developed a single-point diagnostic method to quantitatively characterize magnetotail flapping motions, which has been validated as a robust and efficient approach through applications to numerous flapping events in magnetotail. This method is predicated on the premise that planetary spacecraft velocities are generally much lower than the CS flapping velocity, enabling the unambiguous separation of spatial and temporal variations and thus accurate diagnosis of flapping characteristics. However, near its perihelion, PSP reaches a velocity exceeding 100 km s^{-1} , which is comparable

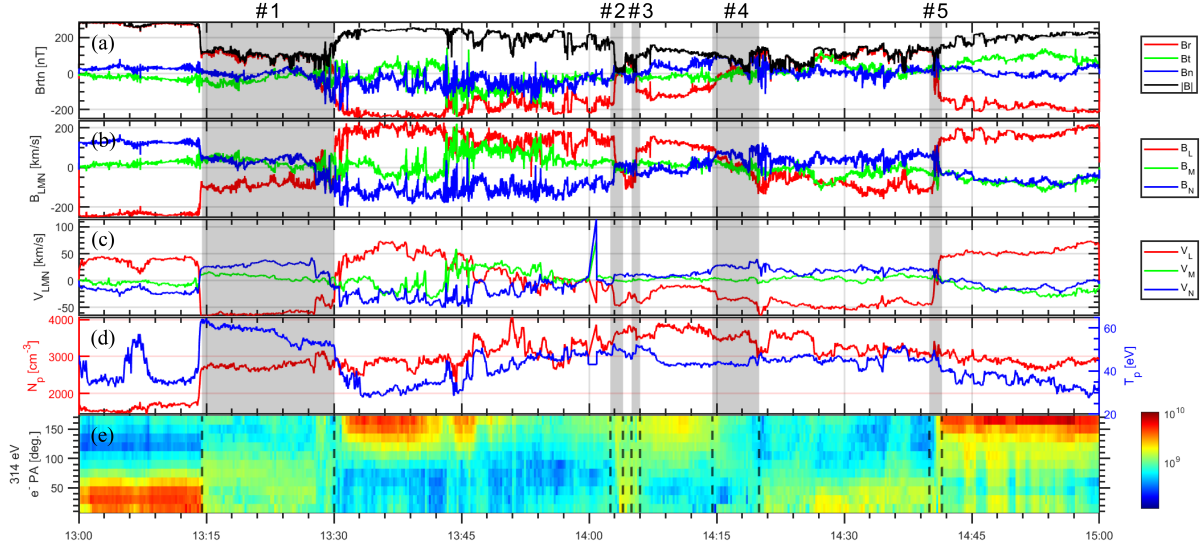


Figure 1. Overview of the event featuring five successive HCS crossings. (a) Components and magnitude of magnetic field in the RTN coordinates. (b) Components of magnetic field in the LMN coordinates of each crossing. (c) Components of proton bulk velocity in the LMN coordinates (minus the average bulk velocity). (d) Proton density and proton temperature. (e) Pitch angle distribution of 314 eV strahl electrons. Magnetic field data is provided by FIELDs aboard PSP, while the solar wind data is provided by SWEAP/SPAN-Ai. Shadow blocks mark the time intervals of the five HCS crossings, which is characterized by the reversal in B_r and concurrent counter-stream/dropout of 314 eV strahls.

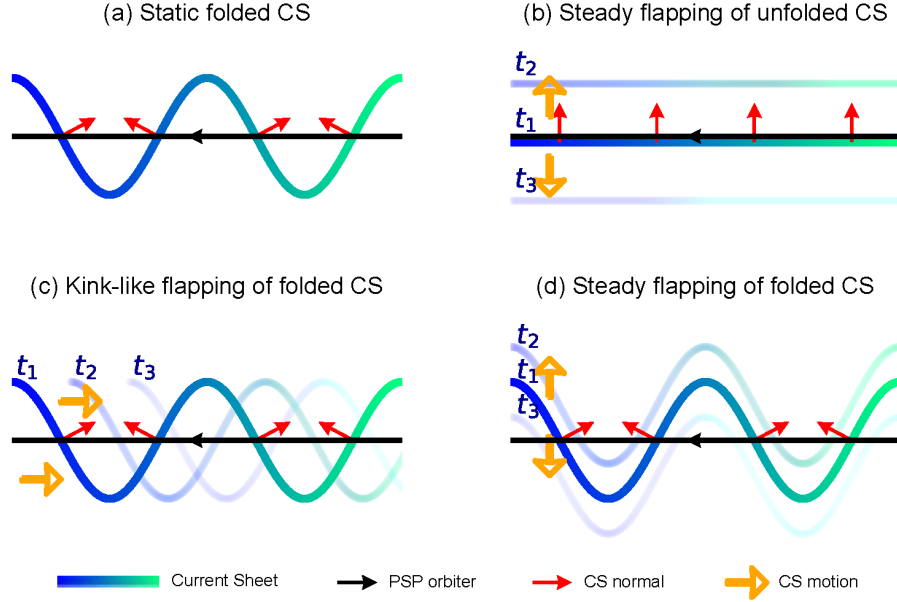


Figure 2. Possible situations responsible for successive CS crossings detected by PSP. (a) CS is folded without flapping motion. (b) CS is unfolded with steady flapping motion. (c) CS is folded with kink-like flapping motion. (d) CS is folded with steady flapping motion. All these sketches are shown in the inertial reference frame. The colored curves in 2D (or surfaces in 3D) represent the HCS, the black solid lines represent the orbit of PSP, the red arrows denote the CS normal direction at each crossing, and the orange arrows indicate the flapping motion of the CS.

Table 1
Details of Each of the Five HCS Crossings

Crossing	MVA Method		CS Normal		Smith Test		$V_{\text{flap},N}$
Index (i)	MVA Time Interval	λ_2/λ_3	e_N in RTN Coordinate	$\langle e_{N,i}, e_{N,i+1} \rangle$	P_1	P_2	(km s ⁻¹)
1	13:29:30–13:31:00	5.22	(−0.35, −0.93, −0.08)	142°	0.131	0.600	−119.8
2	14:02:00–14:04:00	8.97	(−0.17, 0.93, −0.33)	48°	0.176	0.374	62.2
3	14:05:20–14:06:10	4.74	(0.04, −0.39, 0.92)	89°	0.193	0.840	5.2
4	14:14:05–14:18:50	11.8	(−0.32, 0.88, 0.36)	135°	0.186	0.678	−127.4
5	14:40:20–14:40:50	9.26	(0.03, −0.94, −0.35)	...	0.130	0.626	14.0

to the HCS flapping velocity in our case. Consequently, PSP's trajectory during the observation window may exceed the spatial variation scale of the HCS. This caveat prevents us from distinguishing whether the observed wavy HCS structure arises from the intrinsic small-scale folded morphology of the HCS or from propagating flapping waves, and it is not feasible to directly derive the flapping velocity vector of the near-Sun HCS. Instead, we can obtain the velocity component along the local HCS normal direction ($\mathbf{e}_N$) by analyzing the discontinuity type of each of these five HCS crossings.

According to the magnetic field change characteristics across a discontinuity, E. J. Smith (1973) defined two dimensionless parameters to classify discontinuity types, $P_1 = \mathbf{B}_n/|\mathbf{B}|_L$ and $P_2 = \delta|\mathbf{B}|/|\mathbf{B}|_L$. Here, $\mathbf{B}_n$ denotes the normal component of the magnetic field, $|\mathbf{B}|_L$ is the larger one of the field magnitudes on either side of the discontinuity, and $\delta|\mathbf{B}|$ is the absolute difference between these two magnitudes. The classification criteria given by E. J. Smith (1973) are as follows:

$$P_1 > 0.4 \ \& \ P_2 < 0.22 \\ \rightarrow \text{rotational discontinuity (RD);} \quad (1)$$

$$P_1 < 0.22 \ \& \ P_2 > 0.22 \\ \rightarrow \text{tangential discontinuity (TD).} \quad (2)$$

B. T. Tsurutani et al. (1996) proposed the remaining two discontinuity types with slight adjustments to the thresholds:

$$P_1 > 0.2 \ \& \ P_2 > 0.2 \\ \rightarrow \text{neither discontinuity (ND);} \quad (3)$$

$$P_1 < 0.4 \ \& \ P_2 < 0.2 \\ \rightarrow \text{either discontinuity (ED).} \quad (4)$$

As shown in Table 1, we obtain P_1 and P_2 for each HCS crossing and identify all of them as tangential discontinuities. A key characteristic of tangential discontinuities is the absence of mass interchange across the boundary; specifically, in the frame of reference of the TD, there is no plasma flow in the normal direction. This implies that the solar wind velocity component along the HCS normal ($\mathbf{e}_N$) directly corresponds to the flapping velocity of the HCS itself. We take the mean value of the solar wind velocity during each crossing as the flapping velocity (projected onto the HCS normal direction), as listed in the rightmost column of Table 1. These flapping velocities range from several to hundreds of kilometers per second—an appreciable magnitude indicating that the HCS undergoes flapping motion with non-negligible velocity.

4. Origins: Local Generation versus Solar Source

A key question arises: what mechanisms drive the formation of multiple folds in the HCS and its concurrent flapping motion? Could these features originate from local solar wind conditions, such as the KHI, or are they related to solar activities occurring in their source regions? To address this, we analyze in situ solar wind measurements from PSP and remote-sensing data of the solar atmosphere from Solar Dynamics Observatory/AIA, respectively.

4.1. Origin of Multiple Folds

The KHI is a fluid instability that occurs at the interface between two flows with different velocities. In the classical hydrodynamic framework, it is triggered as long as there is a

Table 2
Comparison of the Observed Velocity Shear and Threshold of KHI

Crossing Index	1	4	5
v_{shear}	23.69	6.05	5.65
$\Delta v_{\text{threshold}}$	23.80	14.64	16.32

velocity shear (S. Chandrasekhar 1961). In magnetized plasmas, however, there is a threshold for the velocity shear to overcome the magnetic tension to initiate KHI. The instability criterion proposed by R. Smets et al. (2002) is expressed as

$$(\mathbf{k}_T \cdot \Delta \mathbf{v})^2 > \left(\frac{1}{\rho_1} + \frac{1}{\rho_2} \right) [\rho_1 (\mathbf{k}_T \cdot \mathbf{v}_{A1})^2 + \rho_2 (\mathbf{k}_T \cdot \mathbf{v}_{A2})^2]. \quad (5)$$

Here, $\Delta \mathbf{v} = \mathbf{v}_1 - \mathbf{v}_2$ represents the velocity shear, v_A represents the local Alfvén speed, and $\mathbf{k}_T$ represents the tangential component of wavevector, i.e., the projection of wavevector in L - M plane. Considering that $\mathbf{e}_L$ consistently slopes in the R direction, KHI produced by the velocity shear along $\mathbf{e}_L$ will cause the HCS normal to swing in the R direction, which is consistent with our findings shown in Table 1. Therefore, we focus exclusively on the velocity shear along $\mathbf{e}_M$. In addition, partial crossings are not taken into account because they only capture solar wind conditions on one side of the HCS. Due to the indeterminacy of the wavevector direction, we calculate the threshold of Δv as a function of wavevector orientation. We define v_{shear} as the actual velocity shear along the $\mathbf{e}_M$ direction, and $\Delta v_{\text{threshold}}$ as the minimum velocity shear required to trigger the KH instability (i.e., the critical velocity shear under the most favorable $\mathbf{k}_T$ conditions). A comparison between v_{shear} and $\Delta v_{\text{threshold}}$ is presented in Table 2. Our results show that across all possible wavevector directions, the corresponding Δv thresholds are consistently lower than the observed velocity shear magnitudes. This indicates that even under the most favorable wavevector orientation for KHI initiation, the actual velocity shear is insufficient to trigger the instability. Therefore, we confirm that the multiple folds of HCS are not generated by local KHI.

4.2. Origin of Flapping Motion

To locate the source region of the in situ observation area of PSP, we employ a two-step ballistic backmapping method: first, tracing along the Parker spiral back to the footpoint on the source surface ($2.5 R_\odot$), and then tracing further back to the photosphere based on the potential field source surface (PFSS) model. The footpoint on the photosphere is marked in the 193 Å global image of Carrington rotation (CR) 2239, as illustrated in Figure 3. We find that the footpoint is located near an active region, which is designated as AR12796 by the National Oceanic and Atmospheric Administration.

It is particularly intriguing to investigate the activity within this active region prior to our in situ observation of HCS flapping. Our analysis reveals that a solar flare erupted within this active region between 03:23 and 03:37 UTC on the observation day, triggering intense oscillations in nearby coronal loops. To characterize these dynamics, we select three slits across the loops, corresponding to the three rectangles marked in Figure 4(a), and observe their behaviors before and after the flare eruption. Figure 4(b) shows the temporal evolution of the three slits as observed by AIA 171 Å

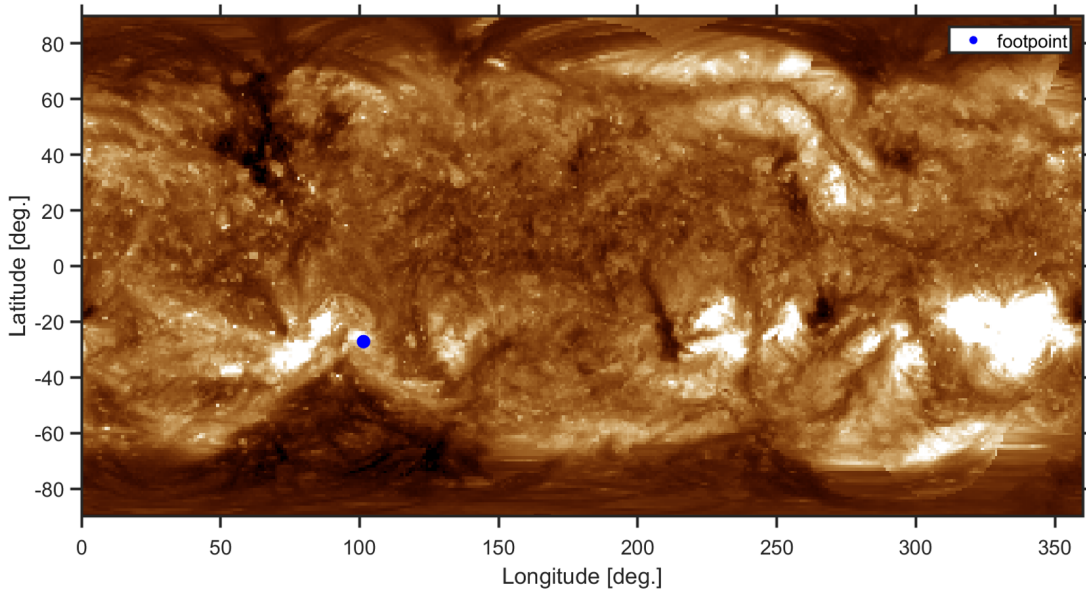


Figure 3. Location of the footpoint on the photosphere. The background is 193 Å global image of CR2239, shown in Carrington coordinates. The blue point represents the resulting footpoint obtained by the two-step ballistic backmapping method, and the bright patch near it is the active region AR12796.

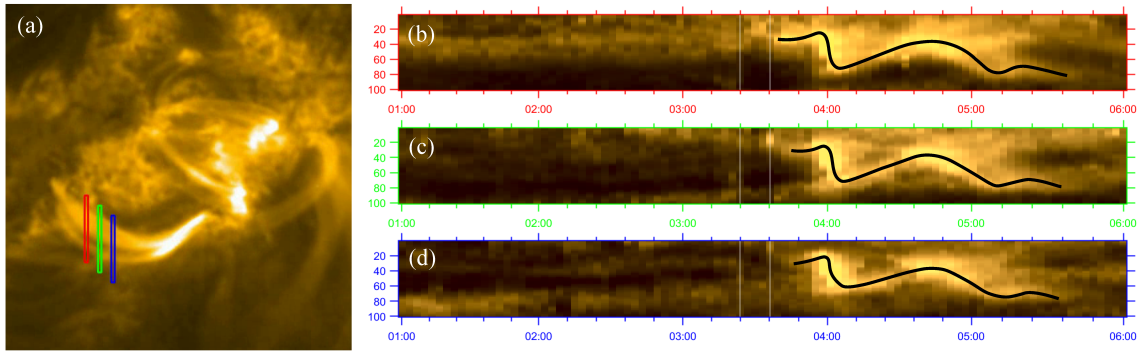


Figure 4. Slit observations of the coronal loop dynamics before and after the solar flare, observed by AIA 171 Å. (a) A close-up view of the active region AR12796 shortly after the flare onset, showing brightened coronal loops. The red, green, and blue narrow rectangles overlaid across the coronal loops represent the selected slits for temporal analysis. (b)–(d) Temporal evolution of coronal loops in the three observing slits. Each panel on the right corresponds to the slit of the same color in panel (a). In each time-series panel, the two white vertical lines mark the beginning and end of the solar flare interval as recorded by Helioviewer.org, while the black solid lines trace the dynamic motion of the coronal loops.

passband, with the motions of coronal loops highlighted by solid lines. One can see that, prior to the flare, these areas remained relatively quiescent and dim. Immediately following the flare eruption, however, these loops exhibit a sudden brightening accompanied by large-amplitude oscillations that persist for over 1.5 hr. This confirms that the flare directly drove the intense oscillation of coronal loops and significantly altered the magnetic field configuration. As these oscillations propagate upward along magnetic field lines into the interplanetary medium, they can induce vibrations in the magnetic field lines themselves. Given that the HCS is a fundamental magnetic field structure, such vibrations would consequently drive its vertical flapping motion.

5. Conclusion and Discussion

To sum up, through the analyses of the five successive crossings of HCS observed by PSP, we point out that the HCS exhibits a complex morphology and dynamic behavior. It is

not only highly curved, forming multiple small-scale folds, but also undergoes flapping motions—oscillating upward and downward at speeds ranging from several to hundreds of kilometers per second.

Figure 5(a) presents the Global Oscillation Network Group synoptic map of CR2239 and the PFSS model solution at the source surface located at $2.5 R_{\odot}$. We find that the projection of trajectory of PSP during this day intersects the magnetic neutral line on the source surface. Based on the modeling of solar corona and heliosphere by Predictive Science Inc. and the Parker solution of solar wind (E. N. Parker 1958), we construct a schematic 3D representation of the global HCS shown in Figure 5(b). PSP traverses the multifold HCS, detecting five successive crossings, while the entire HCS is flapping at the same time. This suggests a far more complex and dynamically evolving heliosphere than previously appreciated.

As for the origins, we compared the observed velocity shear with the threshold of KHI, and verified that these multiple

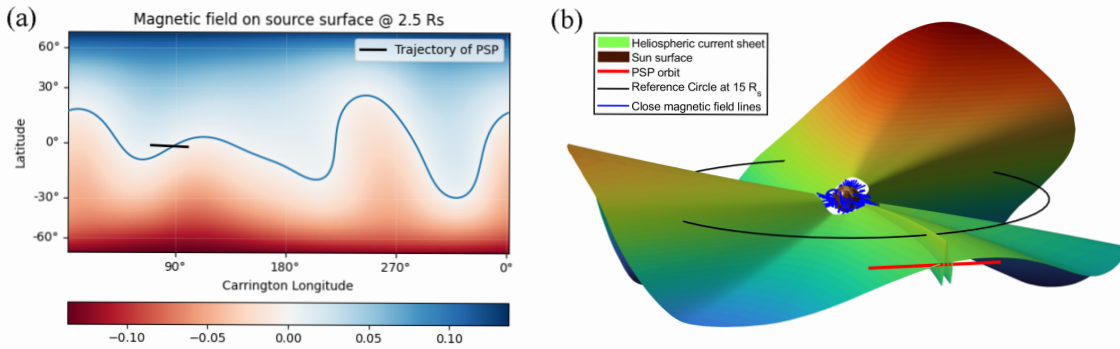


Figure 5. (a) Distribution of magnetic field radical component on the source surface at $2.5 R_{\odot}$. The blue curved line represents the magnetic neutral line with $B_r = 0$, and the black segment represents the projection of PSP’s trajectory during this day. (b) Sketch of the HCS based on the neutral line in panel (a) and the Parker solution. The red segment represents the trajectory of PSP, the collection of blue curved lines represents the closed magnetic field lines beneath the HCS, and the black circle serving as a reference is located at $15 R_{\odot}$ in the ecliptic plane.

folds do not come from KHI locally. Instead, their origins are more likely to lie in the source region, where small structures and activities may slightly modify the HCS neutral line, or in the evolution of solar wind and HMF en route to the PSP location ($20.4 R_{\odot}$). Determining the exact origin requires further investigation of the magnetic footpoint of the PSP observation on the source surface and the intervening heliospheric region. Conventional magnetic field extrapolation models, with their typical spatial resolution, cannot resolve such small-scale structures in a global context. Future high-resolution extrapolation models are expected to distinguish between these two scenarios and provide a definitive answer.

Additionally, we identified a flare erupting near the footpoint on the photosphere, which aligns perfectly with our event in both location and timing. The coronal loops beneath the observed segment of the HCS were driven into strong oscillations, and the upward-propagating oscillations are expected to induce flapping of the HCS itself. Through kinematic simulation, C. Fry & S.-I. Akasofu (1985) suggested that three successive solar flares can cause the flapping motion of HCS near Earth. Our observations provide direct evidence supporting this scenario and demonstrate that, closer to the Sun, even a single flare can be sufficient to drive flapping motions of the HCS.

Based on the time interval between the flare eruption and PSP’s observation, the average propagation speed of these flare-driven transverse oscillations is estimated to be $\sim 400 \text{ km s}^{-1}$. Such perturbations typically propagate outward at a speed equal to the sum of the solar wind flow speed and the local Alfvén speed. Our in situ measurements show that this sum is of the same order of magnitude as the estimated propagation speed. Near the Sun, the lower solar wind speed is compensated by a higher local Alfvén speed, resulting in a combined propagation speed consistent with that at larger heliocentric distances. This consistency qualitatively confirms a temporal causal link between the solar flare and the observed HCS flapping.

An interesting question is whether the flapping motion of the HCS can be detected by coronagraphs. J. Lin et al. (2005) reported apparent deformation and displacement of the helmet streamer surrounding the HCS, driven by a partial-halo coronal mass ejection. The distortion amplitude was comparable to the solar radius and spanned tens of pixels in observations from the Large Angle and Spectrometric Coronagraph C2 and C3 on board the Solar and Heliospheric Observatory (see Figure 4 in

their article). A coronagraph can detect HCS flapping motion only if two criteria are satisfied: (1) the target HCS segment lies near the off-limb within the coronagraph’s field of view; and (2) the flapping amplitude exceeds twice the spatial scale of a single image pixel. Among current coronagraph missions, only STEREO/COR2 meets criterion (1), with each pixel corresponding to a spatial length of 21 Mm. For our study case, the flapping amplitude is estimated to be 0.66–40 Mm, calculated from the HCS normal flapping velocity and the time interval between adjacent crossings. While the small-scale HCS flapping analyzed herein is undetectable with coronagraph observations at 1 au, the emerging “See and Touch” approach (Z. Wu et al. 2025) offers a promising framework for future integrated studies of HCS dynamics. This technique combines remote-sensing data from forward-viewing coronagraphs (e.g., WISPR aboard PSP) with in situ measurements along the spacecraft trajectory. Specifically, WISPR can preimage oscillations in the solar-proximal segment of coronal streamers at a relatively higher spatial resolution; if these oscillations propagate outward along the streamer, they may later be captured by PSP’s in situ instruments as the spacecraft traverses the same region. Such an integrated approach could enable a more comprehensive characterization of HCS flapping propagation and evolution.

Our study focuses on the folding and flapping of the HCS observed by PSP at $\sim 21.5 R_{\odot}$. To trace the radial evolution of these HCS disturbances, in situ observations from multiple spacecraft missions would be invaluable. For instance, the Solar Orbiter (D. Müller et al. 2020) can provide observations at complementary heliocentric distances, while the BepiColombo mission (J. Benkhoff et al. 2021) orbiting Mercury and various satellite constellations near Earth will also have the opportunity to detect HCS folding and flapping when they are exposed to the solar wind beyond the magnetopause.

It is well established that the IMF, particularly its north-south component, is critical to Earth’s space weather. The folding and flapping of the HCS exert a significant influence on the orientation of the IMF upstream of Earth. Enhanced HCS folding or more dynamic flapping motions would increase the frequency of Earth’s traversal across HCS sectors, leading to frequent variations in the IMF direction and consequently more complex space weather conditions. Therefore, diagnosing and understanding the HCS folding/flapping processes, as well as developing forecasting and mitigation

strategies for their associated space weather effects, are of great practical importance.

In this work, we conducted a detailed analysis of the small-scale folds and flapping dynamics of the inner HCS and evaluated whether they originate from local instabilities or solar sources. The HCS, which extends widely in the heliosphere, is a highly bending structure with complex morphology on a small scale and is a highly dynamic structure with significant flapping speeds. Moreover, our results reveal its close connection to solar activity—particularly solar flares—highlighting a clear example of the coupling between the solar atmosphere and the inner heliosphere.

Looking ahead, numerous open questions remain. What is the fundamental origin of the multiple folds in the HCS? Do folds and flapping motions always occur together and share a causal link? Are such features ubiquitous across the entire HCS? Addressing these questions will be essential for advancing our understanding of the 3D dynamics of the HCS, and we look forward to future studies that will shed new light on these mysteries.

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References

- Alfvén, H. 1977, *RvGeo*, **15**, 271
- Benkhoff, J., Murakami, G., Baumjohann, W., et al. 2021, *SSRv*, **217**, 90
- Chandrasekhar, S. 1961, *Hydrodynamic and Hydromagnetic Stability* (Clarendon)
- Crooker, N., Siscoe, G., Shodhan, S., et al. 1993, *JGRA*, **98**, 9371
- DiBraccio, G. A., Dann, J., Espley, J. R., et al. 2017, *JGRA*, **122**, 4308
- Erkaev, N., Semenov, V., & Biernat, H. 2007, *PhRvL*, **99**, 235003
- Erkaev, N., Semenov, V., Kubyshkin, I., Kubyshkina, M., & Biernat, H. 2009, *JGRA*, **114**, A03206
- Forsyth, C., Lester, M., Fear, R., et al. 2009, *AnGeo*, **27**, 2457
- Foullon, C., Lavraud, B., Wardle, N., et al. 2009, *SoPh*, **259**, 389
- Fry, C., & Akasofu, S.-I. 1985, *P&SS*, **33**, 925
- Golovchanskaya, I., & Maltsev, Y. P. 2005, *GeoRL*, **32**, L02102
- Gosling, J., McComas, D., Skoug, R., & Smith, C. 2006, *GeoRL*, **33**, L17102
- Hesse, M., & Birn, J. 2000, *GMS*, **118**, 295
- Hesse, M., Winske, D., Birn, J., & Kuznetsova, M. 1998, *Substorms-4*, **238**, 437
- Huang, J., Liu, Y. C.-M., Klecker, B., & Chen, Y. 2016, *JGRA*, **121**, 19
- Karimabadi, H., Daughton, W., Pritchett, P., & Krauss-Varban, D. 2003, *JGRA*, **108**, 1400
- Lapenta, G., & Brackbill, J. 2002, *PhPl*, **9**, 1544
- Lavraud, B., Fargette, N., Réville, V., et al. 2020, *ApJL*, **894**, L19
- Lemen, J. R., Akin, D. J., Boerner, P. F., et al. 2011, *The solar dynamics observatory* (Springer), **17**
- Lin, J., Ko, Y.-K., Sui, L., et al. 2005, *ApJ*, **622**, 1251
- Liu, Y.-M., Huang, J., Wang, C., et al. 2014, *JGRA*, **119**, 8721
- Malova, H., Zelenyi, L., Popov, V. Y., et al. 2007, *GeoRL*, **34**, L16108
- Müller, D., Cyr, O. S., Zouganelis, I., et al. 2020, *A&A*, **642**, A1
- Müller, D., Nicula, B., Felix, S., et al. 2017, *A&A*, **606**, A10
- Neugebauer, M. 2008, *JGRA*, **113**, A12106
- Parker, E. N. 1958, *ApJ*, **128**, 664
- Peng, J., Liu, Y. C.-M., Huang, J., Klecker, B., & Wang, C. 2019, *JGRA*, **124**, 9814
- Petrukovich, A., Baumjohann, W., Nakamura, R., et al. 2006, *AnGeo*, **24**, 1695
- Pizzo, V. 1994, *JGRA*, **99**, 4185
- Riley, P., Linker, J., & Mikić, Z. 2002, *JGRA*, **107**, 1136
- Rong, Z., Barabash, S., Stenberg, G., et al. 2015a, *JGRA*, **120**, 3462
- Rong, Z., Barabash, S., Stenberg, G., et al. 2015b, *JGRA*, **120**, 5593
- Rong, Z., Shen, C., Petrukovich, A., Wan, W., & Liu, Z. 2010, *P&SS*, **58**, 1215
- Rosenbauer, H., Schwenn, R., Marsch, E., et al. 1977, *JGZG*, **42**, 561
- Runov, A., Angelopoulos, V., Sergeev, V., et al. 2009, *AnGeo*, **27**, 319
- Sergeev, V., Angelopoulos, V., Carlson, C., & Sutcliffe, P. 1998, *JGRA*, **103**, 9177
- Sergeev, V., Runov, A., Baumjohann, W., et al. 2003, *GeoRL*, **30**, 1327
- Sergeev, V., Runov, A., Baumjohann, W., et al. 2004, *GeoRL*, **31**, L05807
- Shen, C., Rong, Z., Li, X., et al. 2008, *AnGeo*, **26**, 3525
- Sitnov, M., Swisdak, M., Guzzdar, P., & Runov, A. 2006, *JGRA*, **111**, A08204
- Smets, R., Delcourt, D., Chanteur, G., & Moore, T. 2002, *AnGeo*, **20**, 757
- Smith, E. J. 1973, *JGR*, **78**, 2054
- Smith, E. J. 2001, *JGRA*, **106**, 15819
- Suess, S., & Hildner, E. 1985, *JGRA*, **90**, 9461
- Szabo, A., Larson, D., Whittlesey, P., et al. 2020, *ApJS*, **246**, 47
- Toichi, T., & Miyazaki, T. 1976, *P&SS*, **24**, 147
- Tsurutani, B. T., Ho, C. M., Sakurai, R., et al. 1996, *A&A*, **316**, 342
- Volwerk, M., Andre, N., Arridge, C., et al. 2013, *AnGeo*, **31**, 817
- Vörös, Z., Yordanova, E., Varsani, A., et al. 2017, *JGRA*, **122**, 11,442
- Wu, Z., He, J., Hou, C., et al. 2025, *ApJS*, **282**, 4
- Xie, Y., Wei, F., Xiang, C., & Feng, X. 2006, *SoPh*, **238**, 377
- Zhang, C., Rong, Z., Gao, J., et al. 2020, *GeoRL*, **47**, e2019GL086011
- Zhang, C., Rong, Z., Zhang, L., et al. 2023, *JGRA*, **128**, e2022JA031232
- Zhao, X., Feng, X., & Wu, C.-C. 2007, *JGRA*, **112**, A06107